Environment Setup

SQL Server and SSMS for the Data Analytics project, on Windows and Mac

The guide gets you to the first query of the project: SQL Server installed and the DataWarehouseAnalytics database in place, restored from the backup or built from the CSV files. With the fixes for the common errors.

1 Choose your setup

Pick the path for your computer and read only its pages. Each path starts on its own page and ends with a green box: when you reach it, your setup is done.

Your computer	Path	Same screens as the videos	Cost	Start
Windows	**SQL Server Express + SSMS**	yes	no cost	page 3
Mac	**Docker** and Visual Studio Code	no, but the same SQL	no cost	page 7
Mac	**Parallels** with Windows	yes, SSMS inside Windows	about 100 dollars a year, 14-day trial	page 9

My advice

- **On Windows:** the first path. It is the one in the videos.
- **On a Mac:** Docker. It costs nothing, it is light, and the SQL is the same.
- **On a Mac, with SSMS exactly as in the videos:** Parallels, if your Mac has 16 GB of memory.

For everyone

- What you get in the materials: page 2
- When something breaks: page 10

2 What you get in the materials

Download the project materials once. You get this folder:

```
sql-data-analytics-project/
├── 00_setup/                    this guide, and the one-click Docker start for Mac
├── 01_datasets/                 the data, as a backup and as CSV files
│   ├── DataWarehouseAnalytics.bak   the database, ready to restore
│   └── csv-files/               the same data, to build the database yourself
├── 02_docs/                     the project roadmap and notes
└── 03_scripts/                  the queries, in the order of the videos
    └── 00_init_database.sql     builds the database from the CSV files
```

The project runs on one database, `DataWarehouseAnalytics`. Its `gold` schema holds the three tables every query uses: `dim_customers`, `dim_products` and `fact_sales`. The scripts are the ones written in the videos, so you can type along or run them.

> The videos show the folders without numbers. `datasets` there is `01_datasets` here, `scripts` is `03_scripts`, and the scripts already use the new names.

Windows

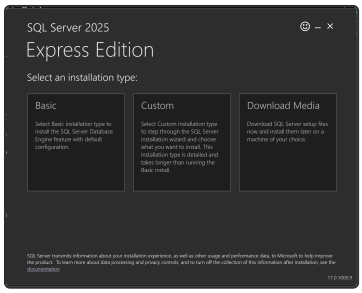
3 Install SQL Server and SSMS

SQL Server Express costs nothing, and it is all you need. The videos show SQL Server 2022. The download page now offers 2025, and it works the same.

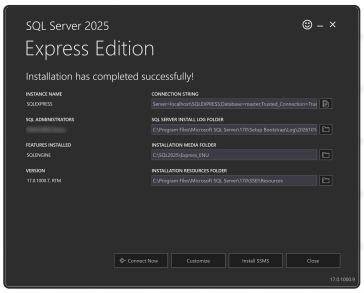
- Download SQL Server Express:
  - Open the SQL Server downloads page.
  - Scroll to the **Express** card.
  - Click **Download now**.
- Run the downloaded file:
  - Choose **Basic**.
  - Accept the licence and click **Install**.
- The last screen shows your server name: `localhost\SQLEXPRESS`. Keep it, you need it in a minute. Your Windows account is now the administrator of this SQL Server.
- Download SQL Server Management Studio (SSMS) and install it:
  - Keep the default options.
  - The installer may offer extra components. You need none of them.



Step 1: the Express card.



Step 2: choose Basic.



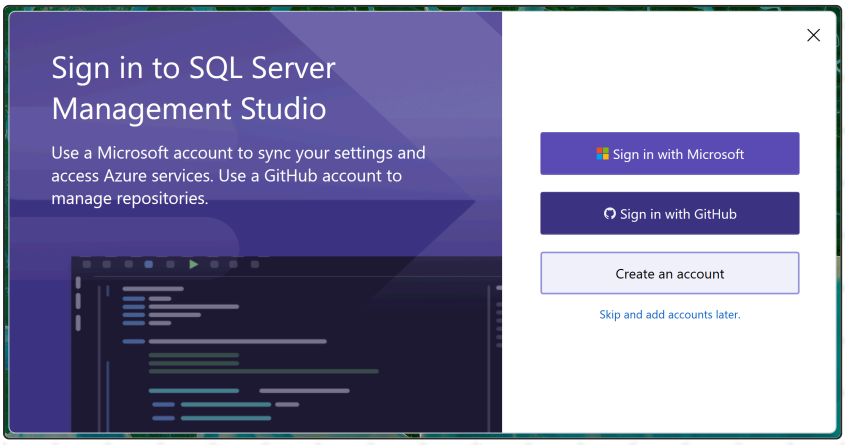
Step 3: the server name.

DATA WITH BARAA

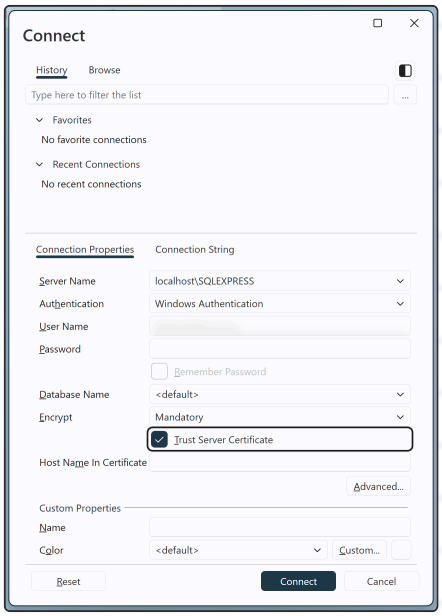
Connect for the first time

SSMS 21 and newer look different from the videos.

- At the first start, SSMS asks you to sign in. You do not need an account: click **Skip and add accounts later** under the buttons.
- In the **Connect** window, fill in three fields:
  - **Server name:** `localhost\SQLEXPRESS` . If the field is empty, that is normal.
  - **Authentication:** Windows Authentication.
  - **Trust Server Certificate:** ticked. Without it you get a certificate error.
- Click **Connect**. Your server appears on the left, in the Object Explorer.



Step 1: the skip link under the buttons.



Step 2: the Connect window, filled in.

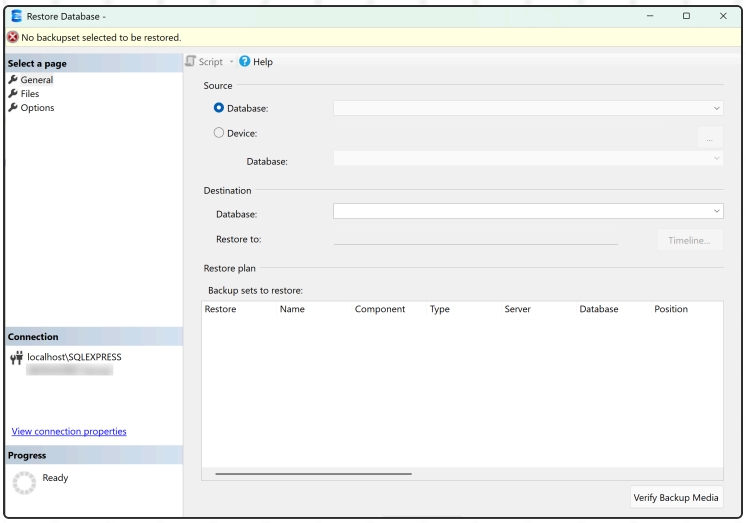
DATA WITH BARAA

4 Create the database

Pick one of the two ways to the same database. Way 1 is a few clicks and needs SQL Server 2022 or newer. Way 2 builds it from the CSV files and works on every version.

Way 1: restore the backup

- Copy `01_datasets/DataWarehouseAnalytics.bak` into the Backup folder of your SQL Server. SQL Server cannot read your Downloads folder, so this copy matters.
  - **SQL Server 2025:** `C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Backup`
  - **SQL Server 2022:** the same path with `MSSQL16`
- In the Object Explorer, right-click **Databases > Restore Database**.
- In the Restore window:
  - Choose **Device** and click the three dots.
  - Click **Add** and pick the backup file.
  - Click **OK** twice.
- Refresh the Databases folder: `DataWarehouseAnalytics` is there.



Way 1: Device, the three dots, Add.

> One rule to know: a backup restores on the same SQL Server version or a newer one, never on an older one. This backup comes from SQL Server 2022. On SQL Server 2019 or older, use way 2.

OR

Way 2: build it from the CSV files

- Create the folder `C:\sql\sql-data-analytics-project` and copy the `01_datasets` folder into it. The CSV files then sit at `C:\sql\sql-data-analytics-project\01_datasets\csv-files\`, the path in the script.
  - **Works:** a folder directly on `C:\`.
  - **Often fails:** Downloads, Desktop, OneDrive. The load then stops with "Access is denied".
  - **Somewhere else:** change the three paths in `00_init_database.sql` to your folder.
- In SSMS, run `03_scripts/00_init_database.sql`:
  - Click **File > Open > File** and pick the script.
  - Click **Execute**, or press F5. It creates the database, the `gold` schema and the three tables, and loads them from the CSV files.

Then, for every script

The project scripts run inside `DataWarehouseAnalytics`. Before you execute, pick it in the database dropdown of the toolbar, where it says `master`. If you built the data warehouse project, you already have this gold layer in `DataWarehouse`, and the queries run there too.

Done. `DataWarehouseAnalytics` is ready. Open `03_scripts/01_database_exploration.sql` and start the project.

Mac with Docker

5 Start SQL Server in Docker

SQL Server does not install on macOS, and SSMS is Windows only. With Docker, SQL Server runs in a small container on your Mac, and you write your queries in Visual Studio Code. The SQL is the same.

Install, once

- Install Docker Desktop, the Apple chip version. Start it and leave it running.
- Open the Docker Desktop settings, **General**. SQL Server ships for Intel chips only, and these two settings let your Mac run it:
  - **Virtual Machine Manager:** Apple Virtualization framework.
  - **Use Rosetta for x86_64/amd64 emulation on Apple Silicon:** on.
- Install the tool you write your queries in:
  - Visual Studio Code.
  - Inside it, the extension SQL Server (mssql) by Microsoft. It is Microsoft's tool for the Mac.

Way 1: one click

- In the project materials, open `00_setup/mac-docker` and double-click `start.command`.
  - **macOS blocks it?** Open System Settings > Privacy & Security, scroll down, click **Open Anyway**, then double-click again. Only the first time.
- Wait a minute. The first start downloads SQL Server. Then the command copies the project files into SQL Server and runs `00_init_database.sql`, which builds `DataWarehouseAnalytics`.
- Nothing to edit: the script runs exactly as in the videos.
- Next time, double-click the same file again.

OR

Way 2: by hand

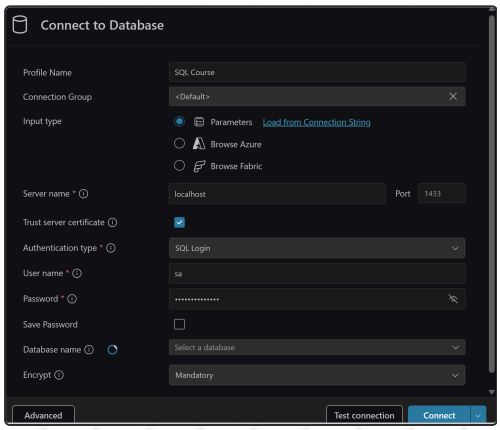
Start SQL Server and copy the project data in. Choose your own password: at least 8 characters with upper case, lower case, a number and a symbol.

```
docker run -e "ACCEPT_EULA=Y" -e "MSSQL_SA_PASSWORD=YourStrong!Passw0rd" \
  -p 1433:1433 --name sql --platform linux/amd64 \
  -d mcr.microsoft.com/mssql/server:2022-latest
docker exec -u root sql mkdir -p /sql/sql-data-analytics-project
docker cp path/to/sql-data-analytics-project/01_datasets sql:/sql/sql-data-analytics-project/
docker exec -u root sql chown -R mssql /sql
```

After you connect, run `03_scripts/00_init_database.sql` from Visual Studio Code. Microsoft's own guide for the first command: SQL Server in a Docker container.

Connect

- In Visual Studio Code, open the SQL Server extension on the left and add a connection:
  - **Server name:** `localhost`
  - **Port:** `1433`
  - **Trust server certificate:** ticked
  - **Authentication type:** SQL Login
  - **User name:** `sa`
  - **Password:** `SqlServer!2026`, or your own from way 2
- Switch to `DataWarehouseAnalytics`: click the database icon at the top right of the editor and pick it.



The connection form, filled in.

Done. `DataWarehouseAnalytics` runs in Docker. Open `03_scripts/01_database_exploration.sql` and start the project.

Mac with Parallels

6 Install Windows in Parallels

This path gives you SSMS exactly as in the videos, because you run Windows on your Mac.

- Install Parallels Desktop. It downloads and installs Windows 11 for you in a few clicks.
  - **Parallels:** about 100 dollars a year, with a 14-day trial.
  - **Windows:** needs its own licence.
- Give the virtual machine memory:
  - **At least:** 6 GB for Windows.
  - **Your Mac:** 16 GB is the comfortable minimum.
- Start Windows in Parallels. From here on, you set up SQL Server like any Windows user.

SQL Server Express installs and runs on the ARM version of Windows 11, even though Microsoft does not list it as supported. Tested with SQL Server 2025 Express and SSMS 22 in Parallels on an Apple chip.

Next: inside Windows, follow the Windows path from page 3.

7 When something breaks

Find your message in the first column. The fix is in the last one.

Install and connect

What you see	Why	What to do
SSMS asks you to sign in at the first start	optional account	Click **Skip and add accounts later**
The server name field is empty	normal	Type `localhost\SQLEXPRESS`
Error 26, "Error Locating Server/Instance Specified"	SQL Server is not running, or the name is wrong	Open **Services**, start "SQL Server (SQLEXPRESS)". If you installed the Developer edition, the server name is just `localhost`
"The certificate chain was issued by an authority that is not trusted"	SSMS encrypts the connection	Tick **Trust Server Certificate**
Installer: "Wait on the Database Engine recovery handle failed"	SQL Server could not start during setup	Uninstall SQL Server, restart Windows, install again. If it fails again, the real reason is in the ERRORLOG file under `C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Log`
SQL Server 2025 setup stops at the rules check	installer problem on some machines	Install SQL Server 2022 Express instead. Everything here runs on it
Your laptop does not let you install anything	no administrator rights	SQL Server needs them. Ask IT, or use a personal computer

The database and the files

What you see	Why	What to do
The restore window does not show the .bak file, or "Access is denied"	SQL Server cannot read Downloads	Copy the file into the Backup folder, section 4, way 1
Msg 3169, the backup "is incompatible with this server"	your SQL Server is older than 2022	Build the database from the CSV files, way 2, or install SQL Server 2022 or 2025 Express
Msg 262, "CREATE DATABASE permission denied in database 'master'"	your Windows login is not an administrator of this SQL Server. Someone else installed it, often IT	Log in with the Windows account that installed SQL Server, or ask an administrator to add your login to the **sysadmin** role (Security > Logins > Properties > Server Roles)
Msg 4860, "Cannot bulk load. The file does not exist or you don't have file access rights"	the file is not at the path in the script	The script reads `C:\sql\sql-data-analytics-project\01_datasets\csv-files\`. Put the folder there, section 4, or change the three paths in `00_init_database.sql`
Msg 4861, "Operating system error code 5 (Access is denied.)"	SQL Server runs as a service and cannot read your folder	Move the `01_datasets` folder to `C:\sql\sql-data-analytics-project\`. Downloads, Desktop and OneDrive folders are the usual culprits

What you see	Why	What to do
Msg 208, "Invalid object name 'gold.fact_sales'"	the query runs in `master`, not in the project database	Switch the database to `DataWarehouseAnalytics` and run again
"'DATETRUNC' is not a recognized built-in function name"	SQL Server older than 2022	Install SQL Server 2022 or newer. Two of the scripts use functions from 2022
A .sql file opens in Notepad, or the .bak shows as text in the browser	the file is not linked to SSMS	Open SSMS first, then **File > Open > File**. For downloads, right-click the link and choose **Save link as**

Docker

What you see	Why	What to do
macOS: "start.command" cannot be opened, or "Apple could not verify" it	the file comes from the internet and is not signed	System Settings > Privacy & Security, click **Open Anyway**, then double-click again
"Cannot connect to the Docker daemon"	Docker Desktop is not running	Start Docker Desktop, wait for the whale icon, try again
"no matching manifest for linux/arm64"	SQL Server is built for Intel chips	Turn on Rosetta in Docker Desktop, section 5. By hand, add `--platform linux/amd64` to the command
"port is already allocated"	another SQL Server listens on 1433	Stop it, or change the first `1433` in `docker-compose.yml` to `1434` and connect to port `1434`
The container stops right after the start	the password breaks the rules	At least 8 characters with upper case, lower case, a number and a symbol
"SQL Server did not answer" from start.command	SQL Server could not start, most often Rosetta is off	Docker Desktop settings, **General**, turn on Rosetta, section 5. The container's log in Docker Desktop names the reason

If you are still stuck, write in the Q&A with the exact error message and a screenshot of the window. Then I can see what happened.